Name: _____

WORKSHEET

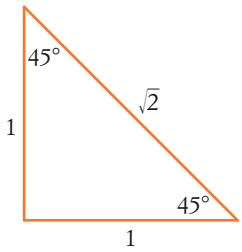
Calculating exact values of trigonometric functions

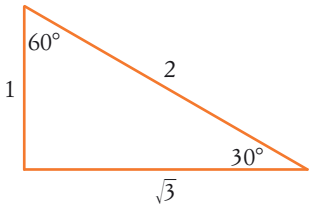
8	12	10	12	1	7	6	10	12	9	5	4	8		

11	8	3	2	4	13	3

Hubert H. Humphrey

Rules





Questions

1 If $\tan(225°) = \tan(180° + \theta) = \tan(\theta)$, what does θ equal?

By simplifying and using the triangles above, calculate the exact value of each of the following.

2 $\tan(225°)$

3 $\cos(120°)$

4 $\sin\left(\frac{5\pi}{3}\right)$

5 $\tan(210°)$

6 $\sin\left(\frac{3\pi}{4}\right)$

7 $\tan\left(\frac{2\pi}{3}\right)$

8 $\cos(330°)$

9 $\tan\left(\frac{7\pi}{3}\right)$

10 $\tan(135°)$

11 $\cos\left(\frac{5\pi}{4}\right)$

12 $\sin(150°)$

13 $\tan\left(\frac{11\pi}{6}\right)$

Solutions

A $-\frac{1}{\sqrt{2}}$

B 1

D $-\frac{1}{\sqrt{3}}$

E $\left(\frac{1}{2}\right)$

G $-\sqrt{3}$

I $\frac{1}{\sqrt{2}}$

N $\frac{\sqrt{3}}{2}$

O $-\frac{\sqrt{3}}{2}$

P $\frac{1}{\sqrt{3}}$

R $45°$

U $\sqrt{3}$

V -1

Y $-\frac{1}{2}$

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